

Speed and velocity

Speed is different from velocity. Speed is how fast you are going and is easy to determine—divide how far you travel by the time it takes. Your velocity is how fast you travel in a particular direction. Changing direction without slowing reduces your velocity, but your speed stays the same.

If you drive 50 miles (80 km)
in two hours, your speed is
25 mph (40 kph).

Accelerating is fun, but defining it in scientific terms can be confusing. This is because acceleration doesn't just mean speeding up. It is any change in velocity. So, it is also used to describe slowing down and changing direction.

The golf ball will keep rolling until friction, gravity, and air resistance slow it down.

Inertia

When things are standing still or moving, they continue to remain in the state they are in (unless force is applied to them to change it). This tendency to be as they are is called inertia.

Turn and learn

Magnetism:
pp. 78-79
Gravity:
pp. 88-89

Balanced forces

Forces act on objects all the time. Opposing forces can be balanced out. When this happens, the object won't be pushed in any direction.

